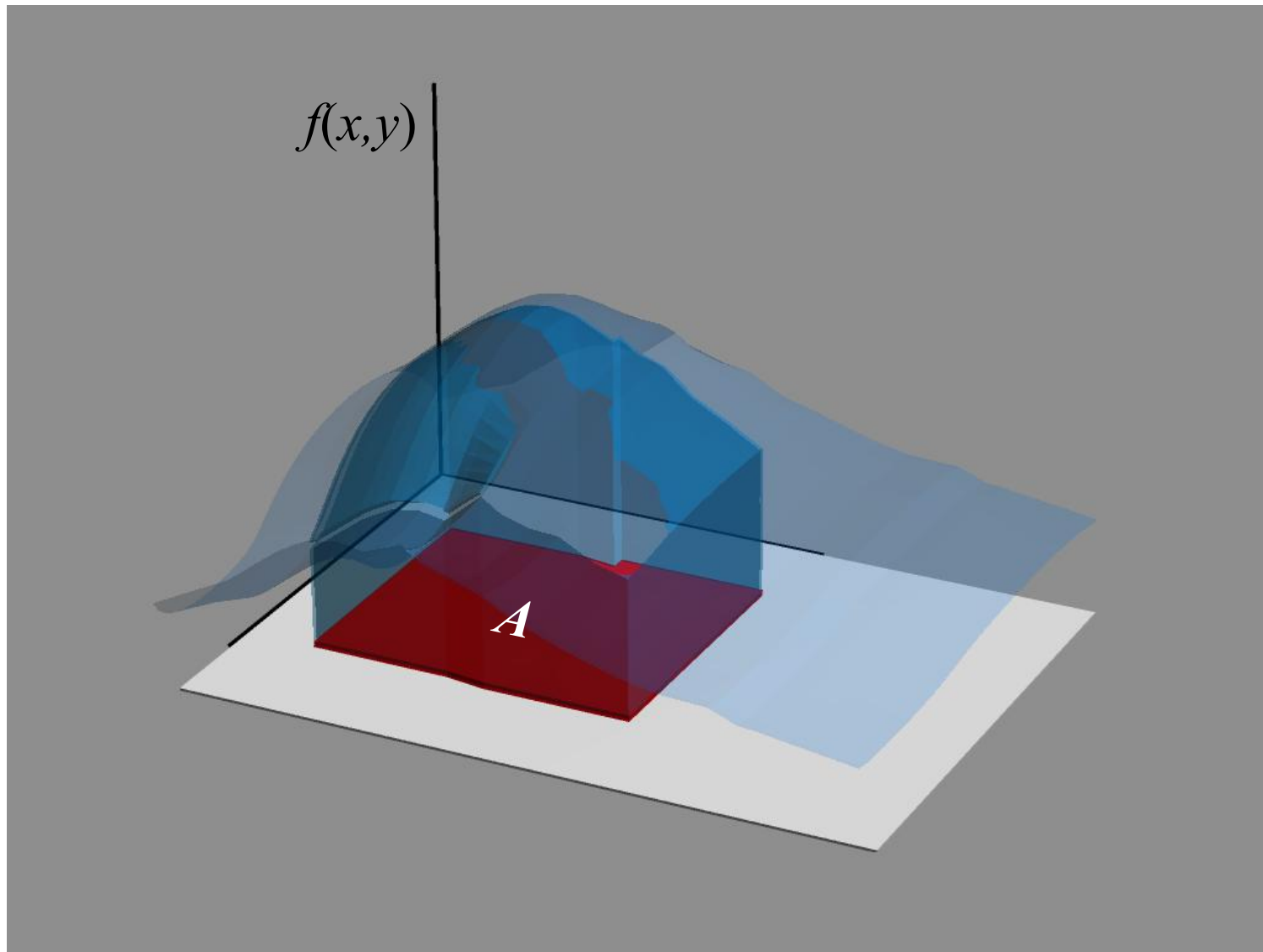
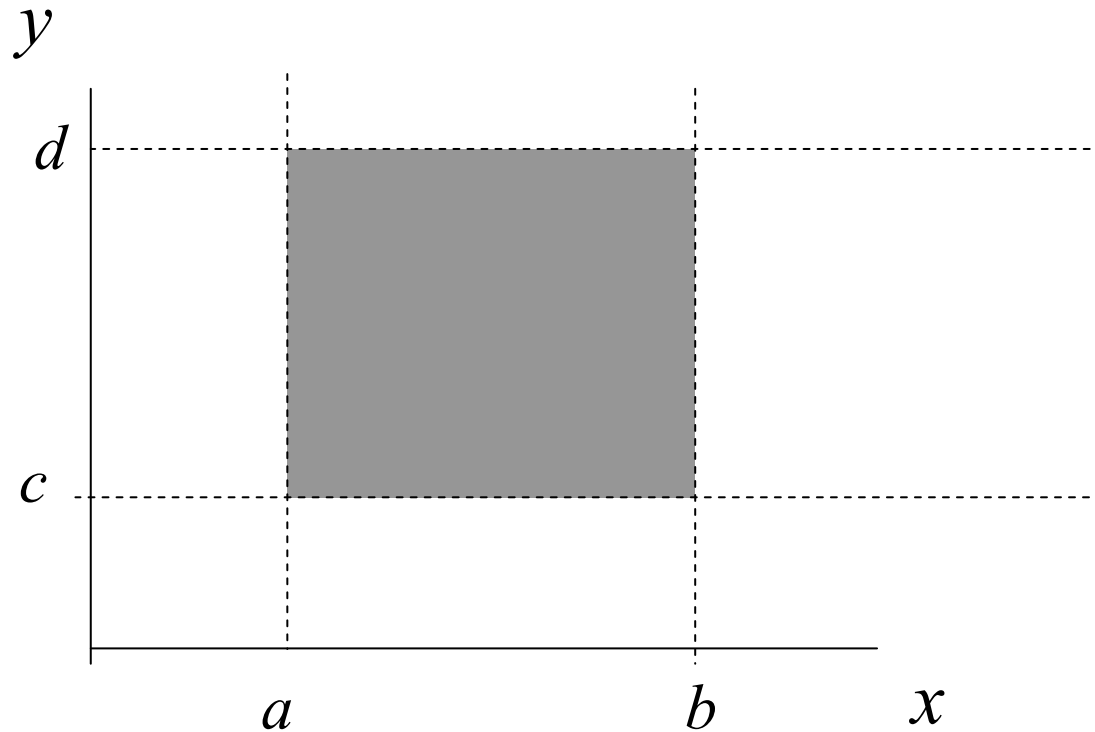


Review of Multiple Integration

$$\iint_A f(x,y) dx dy$$



If the region $A = \{(x,y) \mid a \leq x \leq b, c \leq y \leq d\}$ is a rectangular region with sides parallel to the coordinate axes:



Then

$$\iint_A f(x,y) dx dy = \int_c^d \left[\int_a^b f(x,y) dx \right] dy = \int_a^b \left[\int_c^d f(x,y) dy \right] dx$$

To evaluate

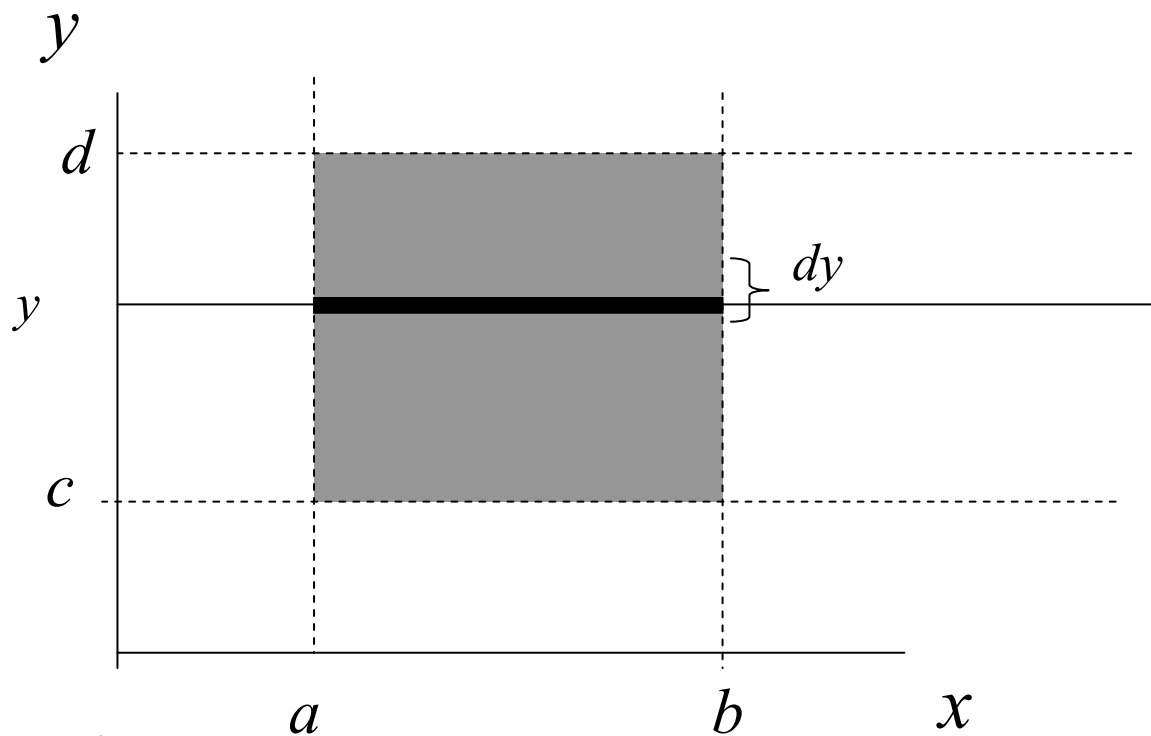
$$\begin{aligned} \int \int_A f(x, y) dx dy &= \int_c^d \int_a^b f(x, y) dx dy \\ &= \int_c^d \left[\int_a^b f(x, y) dx \right] dy \end{aligned}$$

First evaluate the *inner* integral

$$G(y) = \int_a^b f(x, y) dx$$

Then evaluate the *outer* integral

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d G(y) dy$$



$$G(y) = \int_a^b f(x, y) dx = \text{area under surface above the line where } y \text{ is constant}$$

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \underbrace{G(y)}_{\text{Infinitesimal volume under surface above the line where } y \text{ is constant}} dy$$

Infinitesimal volume under surface above the line where y is constant

The same quantity can be calculated by integrating first with respect to y , than x .

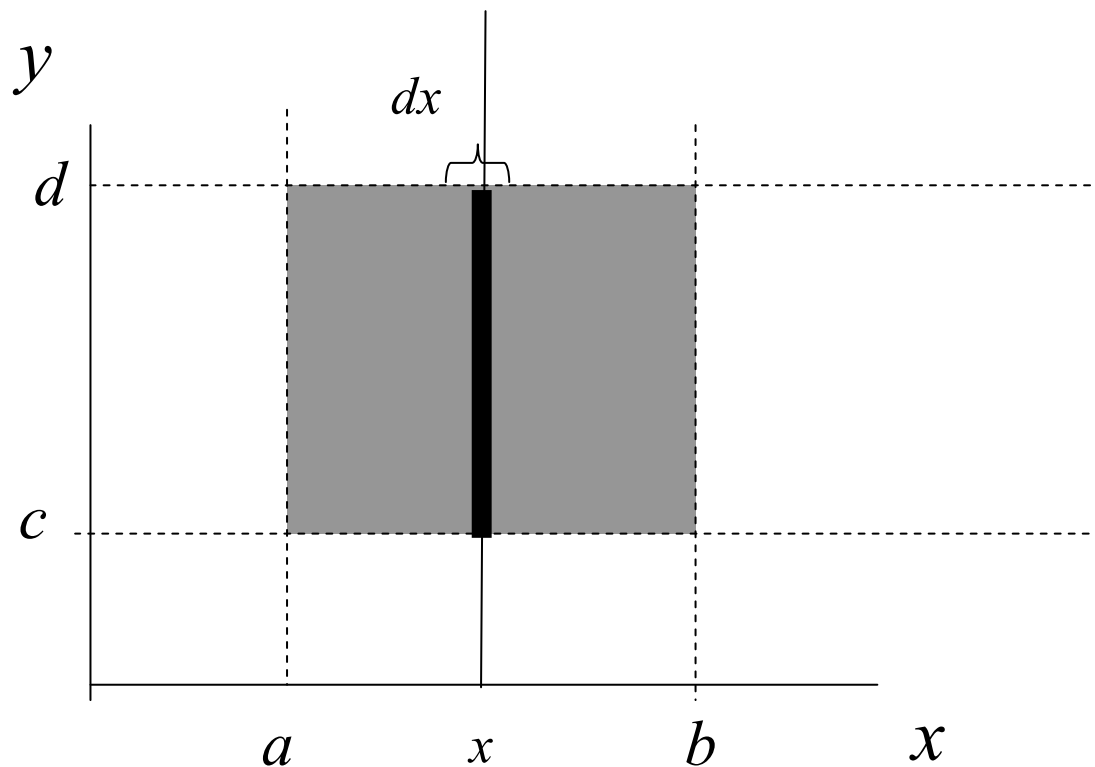
$$\begin{aligned} \int \int_A f(x, y) dx dy &= \int_a^b \int_c^d f(x, y) dy dx \\ &= \int_a^b \left[\int_c^d f(x, y) dy \right] dx \end{aligned}$$

First evaluate the *inner* integral

$$H(x) = \int_c^d f(x, y) dy$$

Then evaluate the *outer* integral

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b H(x) dx$$



$$H(x) = \int_c^d f(x, y) dy = \text{area under surface above the line where } x \text{ is constant}$$

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \underbrace{H(x)}_{\text{Infinitesimal volume under surface above the line where } x \text{ is constant}} dx$$

Infinitesimal volume under surface above the line where x is constant

Example: Compute

$$\int_0^1 \int_0^1 (x^2 y + xy^3) dx dy = \int_0^1 \int_0^1 (x^2 y + xy^3) dy dx$$

Now

$$\begin{aligned} \int_0^1 \int_0^1 (x^2 y + xy^3) dx dy &= \int_0^1 \left[\int_0^1 (x^2 y + xy^3) dx \right] dy \\ &= \int_0^1 \left[\frac{x^3}{3} y + \frac{x^2}{2} y^3 \right]_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{1}{3} y + \frac{1}{2} y^3 \right) dy = \frac{1}{3} \frac{y^2}{2} + \frac{1}{2} \frac{y^4}{4} \Big|_{y=0}^{y=1} \\ &= \frac{1}{6} + \frac{1}{8} = \frac{7}{24} \end{aligned}$$

The same quantity can be computed by reversing the order of integration

$$\int_0^1 \int_0^1 (x^2 y + xy^3) dy dx = \int_0^1 \left[\int_0^1 (x^2 y + xy^3) dy \right] dx$$

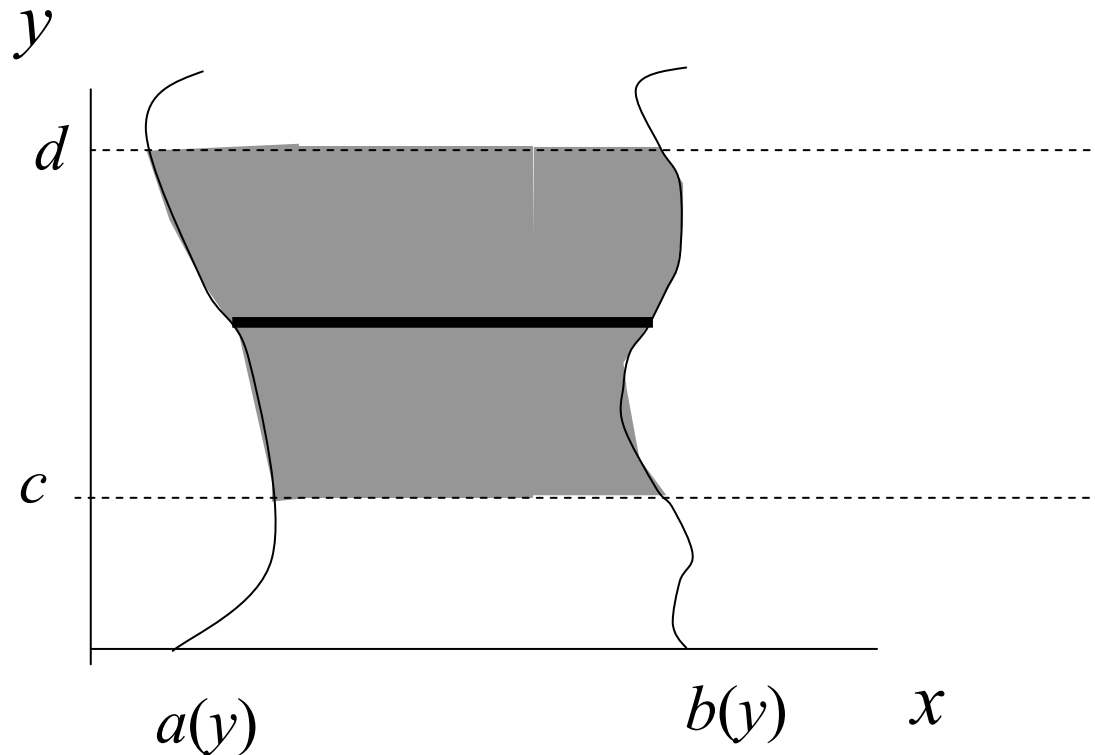
$$= \int_0^1 \left[\frac{y^2}{2} x^2 + \frac{y^4}{4} x \right]_{y=0}^{y=1} dx$$

$$= \int_0^1 \left(\frac{1}{2} x^2 + \frac{1}{4} x \right) dx = \frac{1}{2} \frac{x^3}{3} + \frac{1}{4} \frac{x^2}{2} \bigg|_{x=0}^{x=1}$$
$$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$

Integration over non rectangular regions

Suppose the region A is defined as follows

$$A = \{(x, y) \mid a(y) \leq x \leq b(y), c \leq y \leq d\}$$

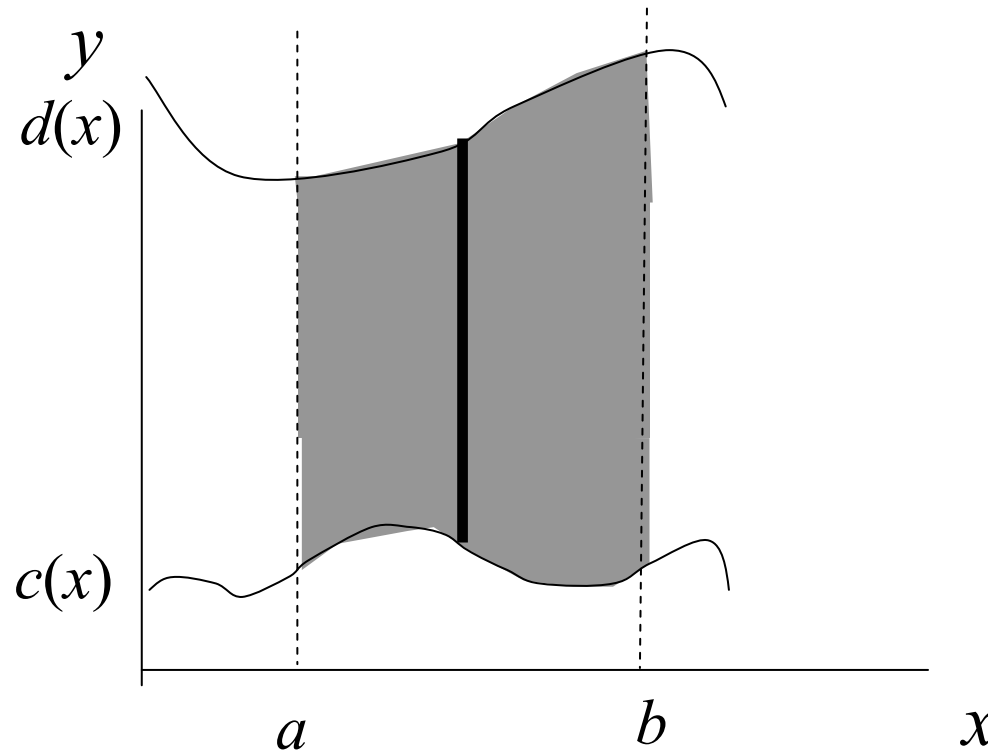


Then

$$\iint_A f(x, y) dx dy = \int_c^d \left[\int_{a(y)}^{b(y)} f(x, y) dx \right] dy$$

If the region A is defined as follows

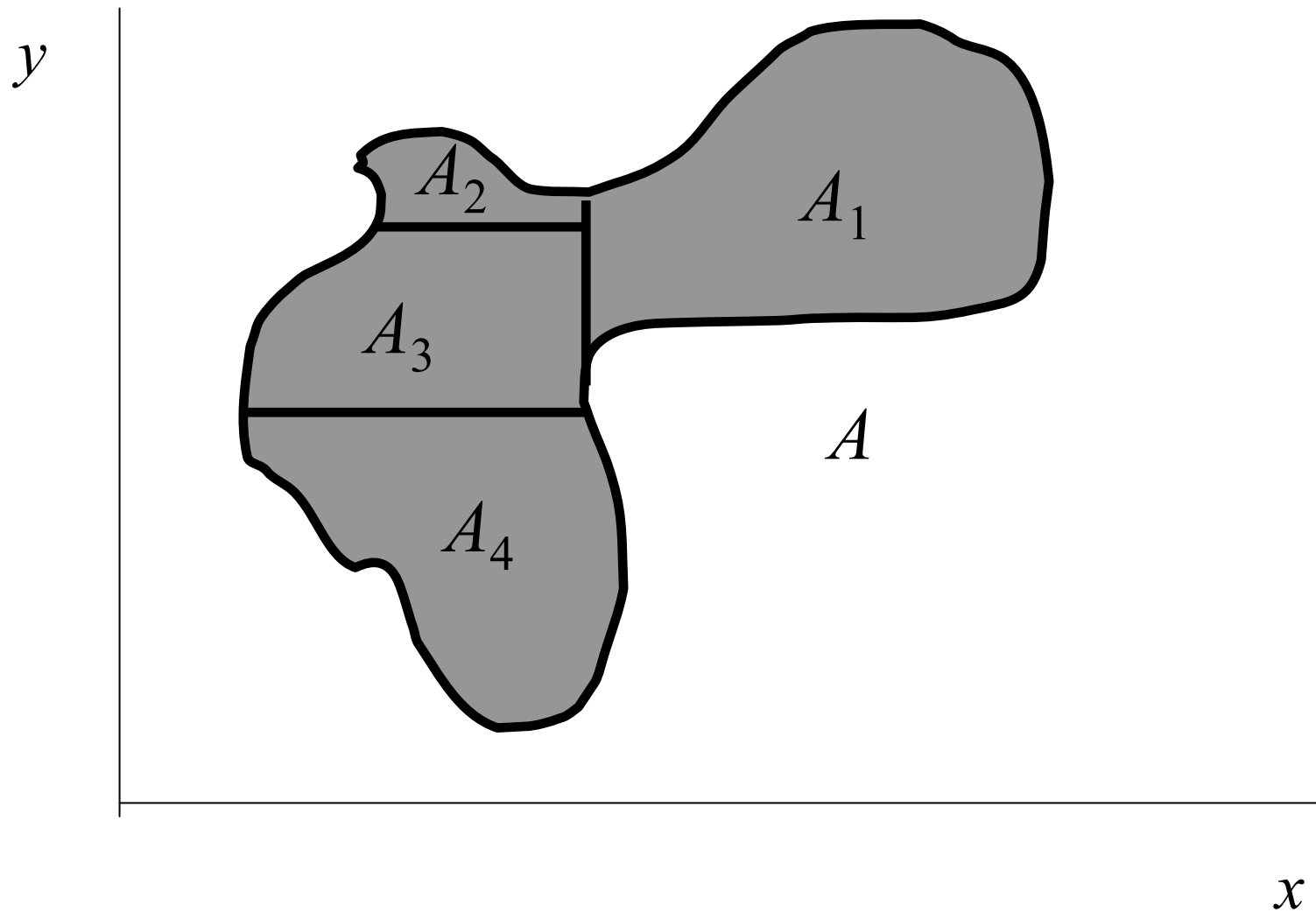
$$A = \{(x, y) \mid a \leq x \leq b, c(x) \leq y \leq d(x)\}$$



Then

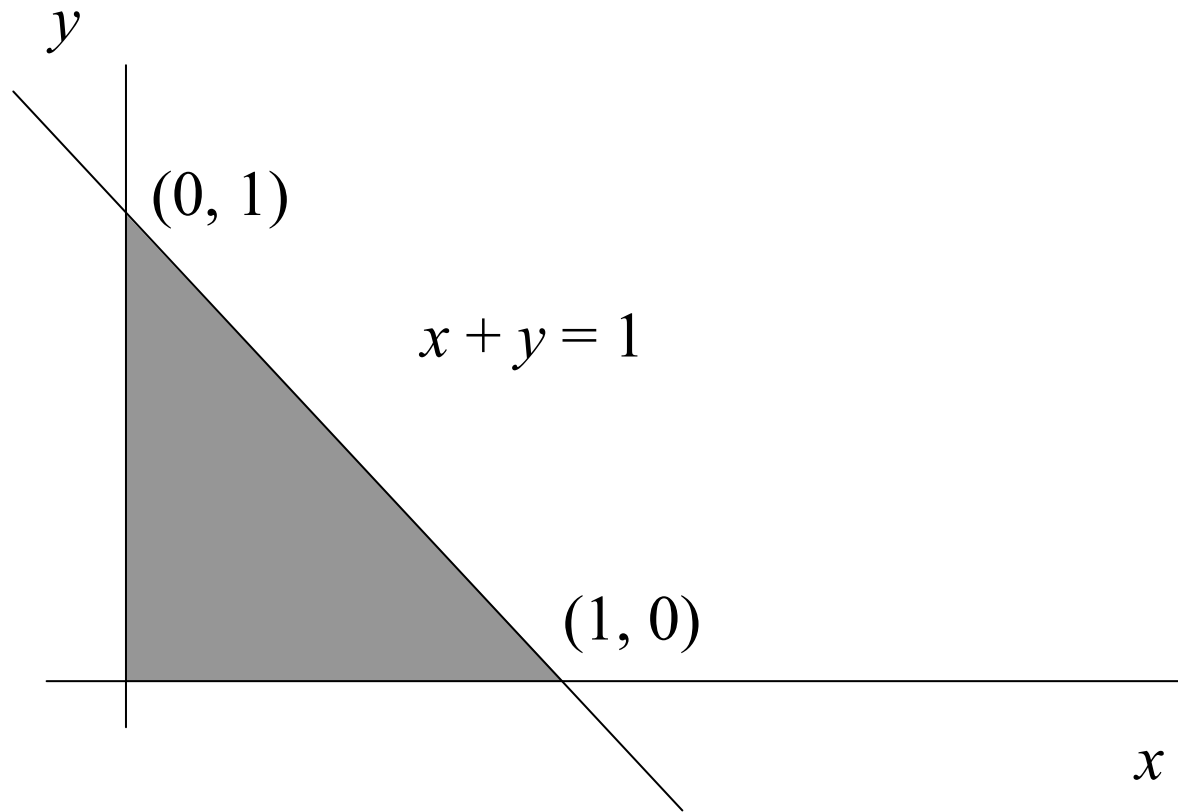
$$\iint_A f(x, y) dx dy = \int_a^b \left[\int_{c(x)}^{d(x)} f(x, y) dy \right] dx$$

In general the region A can be partitioned into regions of either type

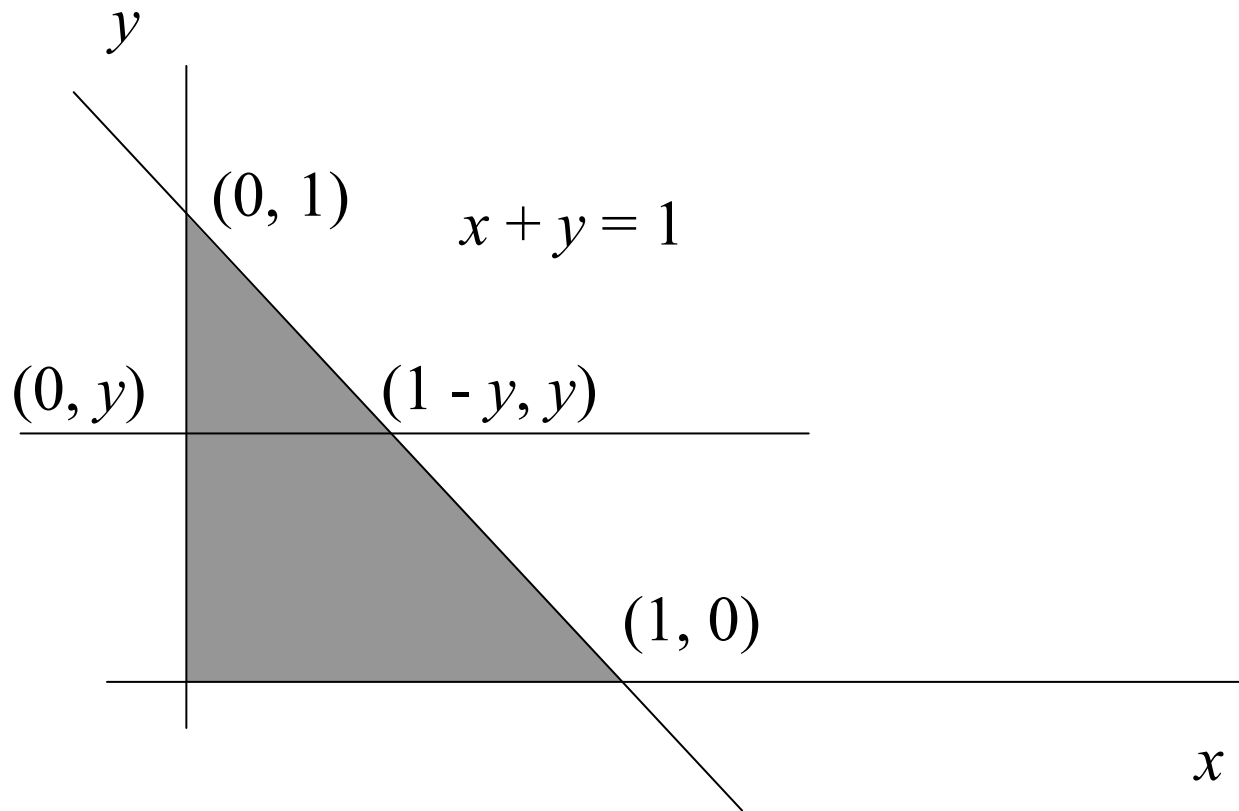


Example:

Compute the volume under $f(x,y) = x^2y + xy^3$ over the region $A = \{(x,y) \mid x + y \leq 1, 0 \leq x, 0 \leq y\}$



Integrating first with respect to x than y

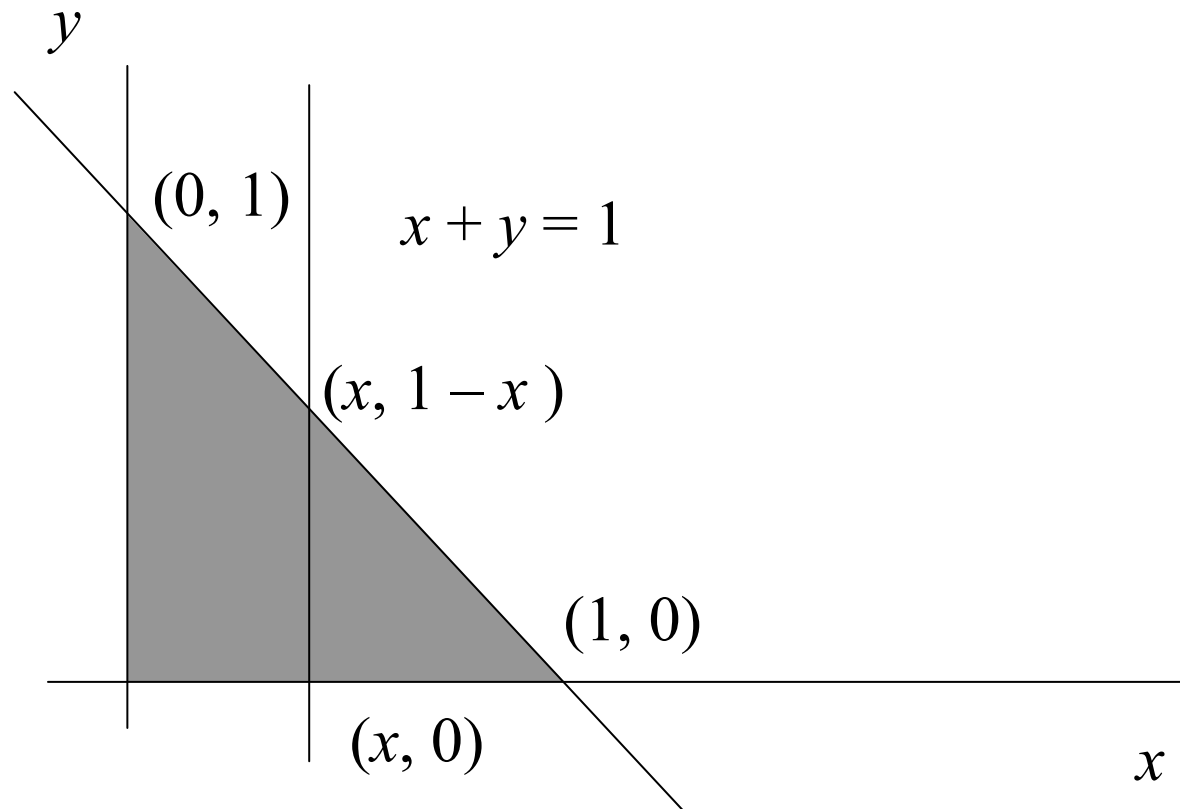


$$\begin{aligned}\iint_A (x^2 y + xy^3) dx dy &= \int_0^1 \int_0^{1-y} (x^2 y + xy^3) dx dy \\ &= \int_0^1 \left[\int_0^{1-y} (x^2 y + xy^3) dx \right] dy\end{aligned}$$

and

$$\begin{aligned} \int_0^1 \left[\int_0^{1-y} (x^2 y + xy^3) dx \right] dy &= \int_0^1 \left[\frac{x^3}{3} y + \frac{x^2}{2} y^3 \right]_{x=0}^{x=1-y} dy \\ &= \int_0^1 \left[\frac{(1-y)^3}{3} y + \frac{(1-y)^2}{2} y^3 \right] dy \\ &= \int_0^1 \left[\frac{y - 3y^2 + 3y^3 - y^4}{3} + \frac{y^3 - 2y^4 + y^5}{2} \right] dy \\ &= \frac{\frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5}}{3} + \frac{\frac{1}{4} - \frac{2}{5} + \frac{1}{6}}{2} \\ &= \frac{1}{6} - \frac{1}{3} + \frac{1}{4} - \frac{1}{15} + \frac{1}{8} - \frac{1}{5} + \frac{1}{12} \\ &= \frac{20-40+30-8+15-24+10}{120} = \frac{3}{120} = \frac{1}{40} \end{aligned}$$

Now integrating first with respect to y than x



$$\begin{aligned} \iint_A (x^2 y + xy^3) dy dx &= \int_0^1 \int_0^{1-x} (x^2 y + xy^3) dy dx \\ &= \int_0^1 \left[\int_0^{1-x} (x^2 y + xy^3) dy \right] dx \end{aligned}$$

Hence

$$\int_0^1 \left[\int_0^{1-x} (x^2 y + xy^3) dy \right] dx = \int_0^1 \left[\left(x^2 \frac{y^2}{2} + x \frac{y^4}{4} \right) \right]_{y=0}^{y=1-x} dx$$

$$= \int_0^1 x^2 \frac{(1-x)^2}{2} + x \frac{(1-x)^4}{4} dx$$

$$= \int_0^1 \frac{x^2 - 2x^3 + x^4}{2} + \frac{x - 4x^2 + 6x^3 - 4x^4 + x^5}{4} dx$$

$$= \int_0^1 \frac{x - 2x^2 + 2x^3 - 2x^4 + x^5}{4} dx$$

$$= \frac{1}{8} - \frac{1}{6} + \frac{1}{8} - \frac{1}{10} + \frac{1}{20} = \frac{15-20+15-12+6}{120} = \frac{4}{120}$$

Joint Continuous Random Vector (Section 5.1, 5.3)

Joint distribution of two cts. R.V.

Definitions:

— Joint P.D.F. =

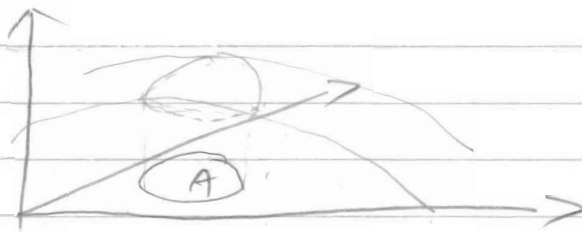
Let X and Y be two continuous R.V. Then a

bivariate function $f(x, y)$ is a joint P.D.F. if

$$1) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1, f(x, y) \geq 0$$

$$2) P((X, Y) \in A) = \iint_A f(x, y) dx dy$$

for all $A \subseteq \mathbb{R}^2$



— Marginal P.D.F.

If we're interested in only the distribution of X or Y . We can find it from the Joint P.D.F. of (X, Y) .

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

They're called marginal P.D.F.

— Conditional P.D.F.

We may be interested in the distribution of X (or Y) given the information of $Y=y$ (or $X=x$). This distribution is characterized by so-called conditional distribution.

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dy}$$

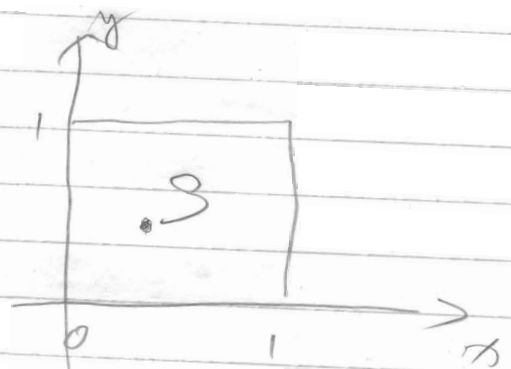
$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx}$$

Example:

$$S = [0, 1] \times [0, 1]$$

$$= \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

$$P(E) = \frac{\text{area}(E)}{\text{area}(S)} = \text{area}(E)$$

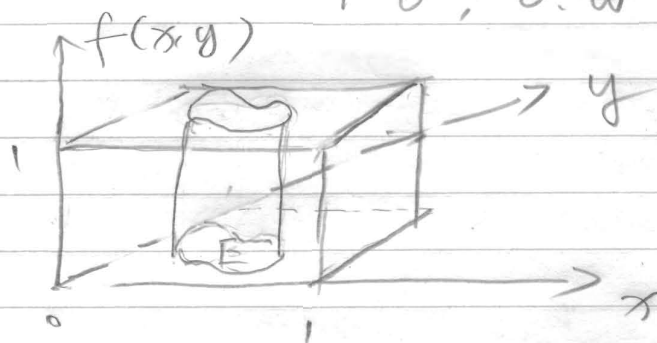


Let x = the x -axis value

y = the y -axis value.

The joint P.D.F. of (X, Y) is given by

$$f(x, y) = \begin{cases} 1, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{o.w} \end{cases}$$



By the definition of double integral we can see that

$$\begin{aligned} \iint_E f(x, y) dx dy &= 1 \times \text{area}(E) \\ &= P(E) \end{aligned}$$

Also $f(x, y) \geq 0$ and $\int_0^1 \int_0^1 f(x, y) dx dy = 1$

Marginal P.D.F.:

$$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 1 dy = 1, \text{ for } x \in [0, 1]$$

$$f_Y(y) = \int_0^1 f(x, y) dx = \int_0^1 1 dx = 1, \text{ for } y \in [0, 1]$$

That is, $X \sim \text{Unif}([0,1])$, $Y \sim \text{Unif}([0,1])$

Conditional P.D.F.

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = 1, \text{ for } x \in [0,1]$$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = 1, \text{ for } y \in [0,1]$$

In this example, we see that

$$f_{X|Y}(x|y) = f_X(x) \text{ for all } x \in [0,1] \\ \text{and } y \in [0,1]$$

$$\text{also, } f_{Y|X}(y|x) = f_Y(y), \text{ for all } x \in [0,1], y \in [0,1]$$

It is also true that

$$f(x,y) = f_X(x) \cdot f_Y(y) \text{ for all } x \in [0,1], y \in [0,1]$$

When this is true, we say that X and Y are independent.

Definition:

Let $f(x,y)$ be the joint P.D.F. of (X,Y) . If

$$f(x,y) = f_X(x) \cdot f_Y(y) \text{ for all } x \in \mathbb{R}, \\ y \in \mathbb{R},$$

then we say X and Y are independent

Example:

A bank has a drive-up facility and a walk-up facility. Let X = the proportion of time that the drive-up facility is used, and

Y = the proportion of time that the walk-up facility is used. The possible values of (X, Y)

is $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Suppose

the joint p.d.f. of (X, Y) is given by:

$$f(x, y) = \begin{cases} \frac{6}{5} (x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

[In the following statistics course, we will

talk about how to "infer" or "estimate"

such a p.d.f. Here, let's assume it and use it]

Marginal p.d.f.

$$f_X(x) = \int_0^1 f(x, y) dy$$

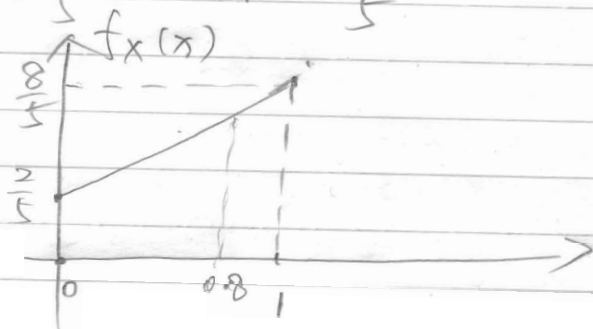
$$= \int_0^1 \frac{6}{5} (x+y^2) dy$$

$$= \frac{6}{5} \times \left(\int_0^1 x dy + \int_0^1 y^2 dy \right)$$

$$= \frac{6}{5} \times \left(x \cdot 1 + \frac{y^3}{3} \Big|_0^1 \right)$$

$$= \frac{6}{5} \times \left(x + \frac{1}{3} \right)$$

$$= \frac{6}{5} x + \frac{2}{5}$$



$$P(X > 0.8) = \int_{0.8}^1 \left(\frac{6}{5}x + \frac{2}{5} \right) dx$$

$$= \frac{1.48}{5}$$

From this P.D.F, we can see the the drive-up facility is more likely to be full than empty.

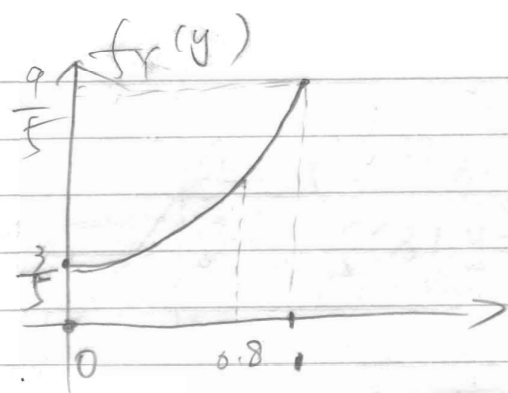
$$f_Y(y) = \int_0^1 f(x, y) dx$$

$$= \int_0^1 \frac{6}{5} (x+y^2) dx$$

$$= \frac{6}{5} \times \left(\int_0^1 x dx + \int_0^1 y^2 dx \right)$$

$$= \frac{6}{5} \times \left(\frac{x^2}{2} \Big|_0^1 + y^2 x \Big|_0^1 \right)$$

$$= \frac{6}{5} \times \left(\frac{1}{2} + y^2 \right) = \frac{3}{5} + \frac{6}{5} y^2$$



$$\begin{aligned}
 P(X > 0.8) &= \int_{0.8}^1 \left(\frac{3}{5} + \frac{6}{5} y^2 \right) dy \\
 &= \frac{1.576}{5}
 \end{aligned}$$

The distribution of Y is similar to that of X . But there are more days of being fairly full.

Let's look at the relationships btw X and Y .

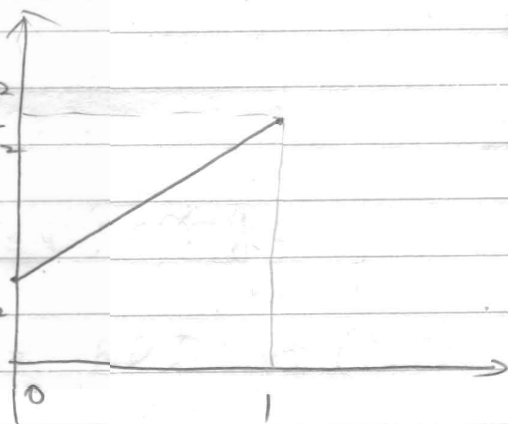
Conditional P.D.F. of X given Y :

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

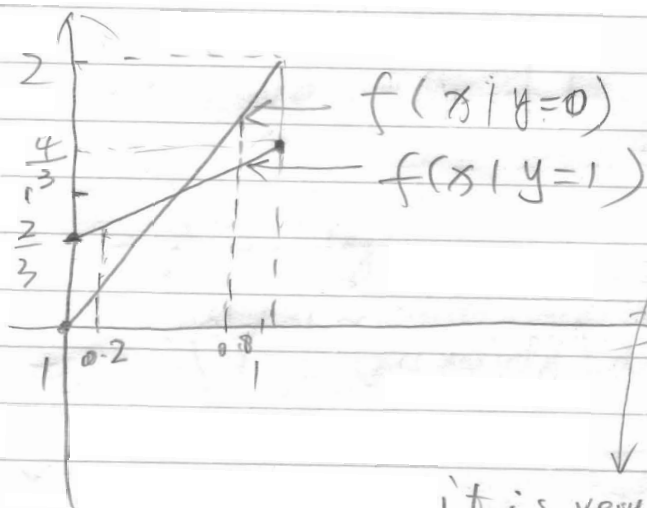
$$= \frac{\frac{6}{5}(x+y^2)}{\frac{3}{5} + \frac{6}{5}y^2}$$

$$= \frac{2(x+y^2)}{1+2y^2}$$

$$= \frac{2y^2}{1+2y^2} + \frac{2}{1+2y^2} X$$



We see that when y is smaller, X is more likely to be large.



When $y=0$

$p(X > 0.8 | Y=0)$ long

$p(X < 0.2 | Y=0)$ small

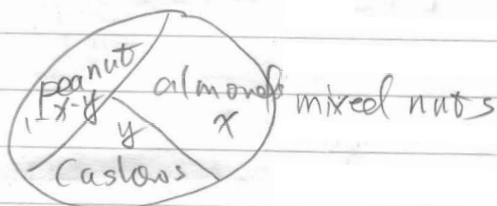
When $y=1$

$p(X > 0.8 | Y=1)$ long

$p(X < 0.2 | Y=1)$ medium

it is very unlikely both X and Y small.
 X and Y are not independent in this example.

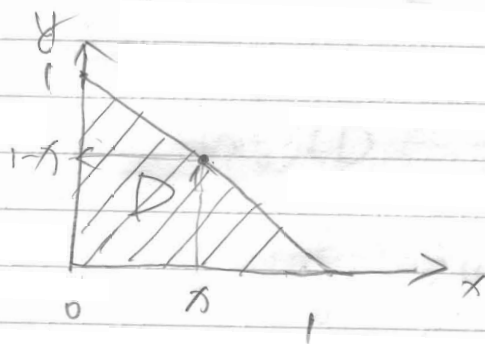
Example



X : weight of almonds

Y : weight of cashews

$$D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq x+y \leq 1\}$$



The joint P.D.F. of (X, Y) :

$$f(x, y) = \begin{cases} 24xy, & 0 \leq x \leq 1, \\ & 0 \leq y \leq 1-x \\ 0 & \text{otherwise} \end{cases}$$

First, let's verify $f(x, y)$ is a joint P.D.F.

$$\iint_D 24xy \, dx \, dy$$
$$= \int_0^1 \left(\int_0^{1-x} (24xy) \, dy \right) dx$$

$$= \int_0^1 \left(24x \cdot \frac{y^2}{2} \Big|_0^{1-x} \right) dx$$

$$= \int_0^1 12x \cdot (1-x)^2 \, dx$$

$$= 12 \cdot \int_0^1 x(1-2x+x^2) \, dx$$

$$= 12 \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right)$$

$$= 1$$

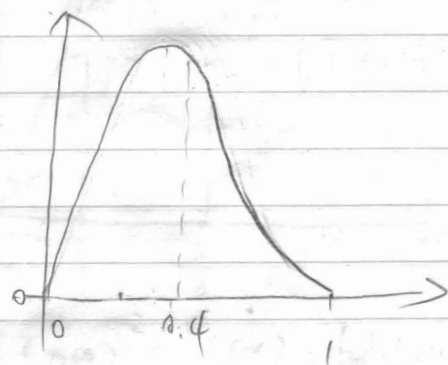
Marginal P.D.F. :

$$f_x(x) = \int_{(x,y) \in D} f(x,y) \, dy$$

$$= \int_0^{1-x} 24xy \, dy$$

$$= 24x \cdot \frac{(1-x)^2}{2} = 12x(1-x)^2$$

for $x \in [0, 1]$.



$$f_Y(y) = \int_{(x,y) \in D} f(x,y) dx$$

$$= \int_0^{1-y} 24xy dx$$

$$= 24y \left. \frac{x^2}{2} \right|_0^{1-y}$$

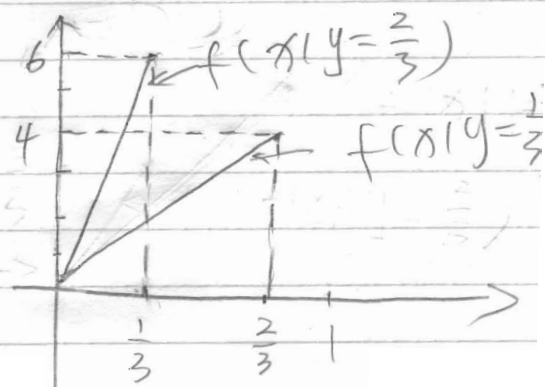
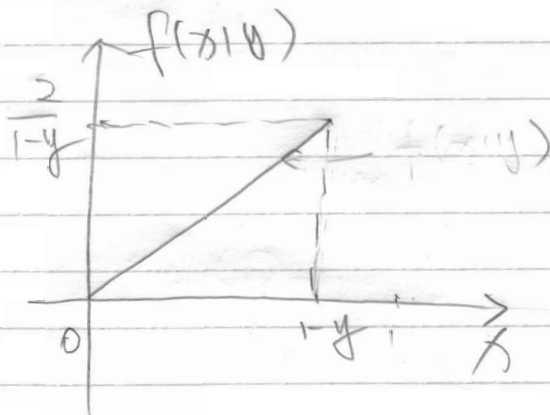
$$= 12y(1-y)^2$$

Conditional P.D.F.

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

$$= \frac{24xy}{12y(1-y)^2}, \quad \text{for } (x,y) \in D$$

$$= \frac{2}{(1-y)^2} x, \quad \text{for } 0 \leq x \leq 1-y$$

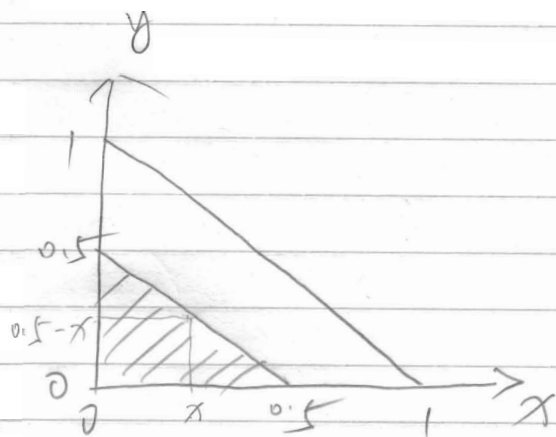


X and Y are not indep. since $f(x,y) \neq f_X(x) \cdot f_Y(y)$
or $f(x|y) \neq f_X(x)$

Computing probability =

$$P(X+Y \leq 0.5)$$

$$= \iint_{\substack{(x,y) \in D \\ x+y \leq 0.5}} f(x,y) dx dy$$



$$= \int_0^{0.5} \left(\int_0^{0.5-x} 24xy dy \right) dx$$

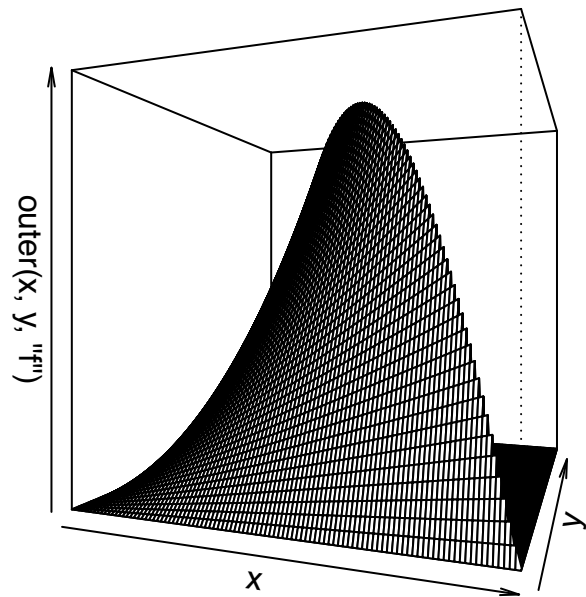
$$= \int_0^{0.5} \left(24x \cdot \frac{(0.5-x)^2}{2} \right) dx$$

$$= \int_0^{0.5} 12x (0.5-x)^2 dx$$

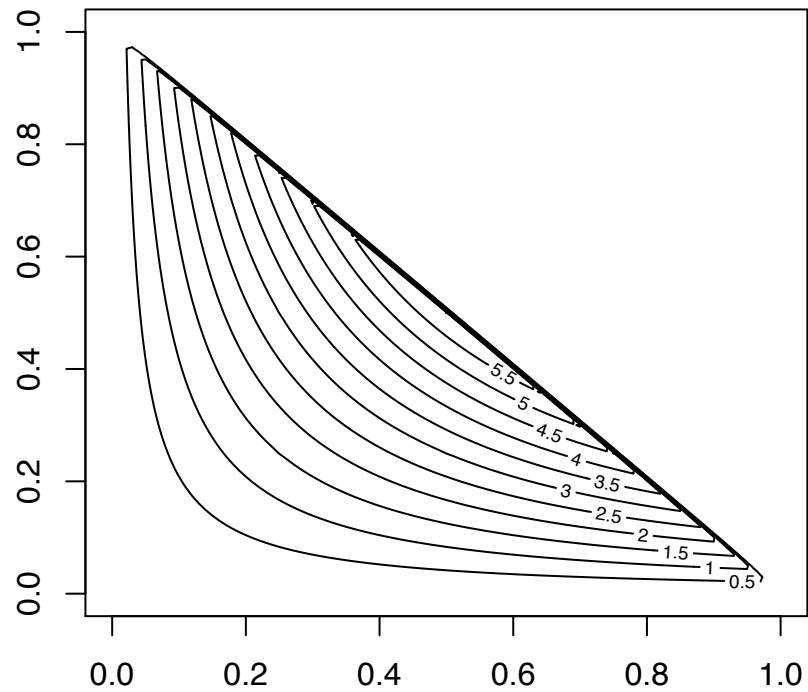
$$= \int_0^{0.5} 12x \left(\frac{1}{4} - x + x^2 \right) dx$$

$$= 0.0625$$

3D PDF

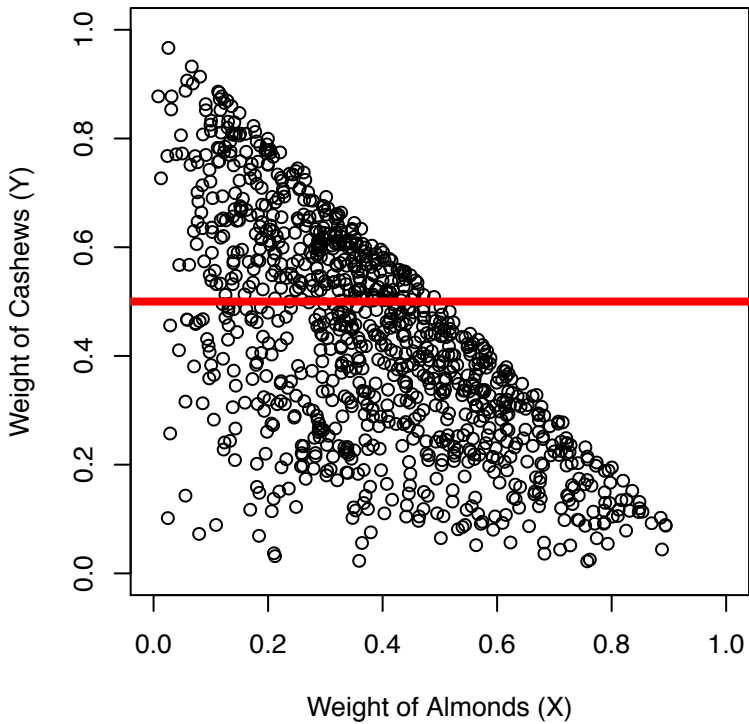


Contour Plot

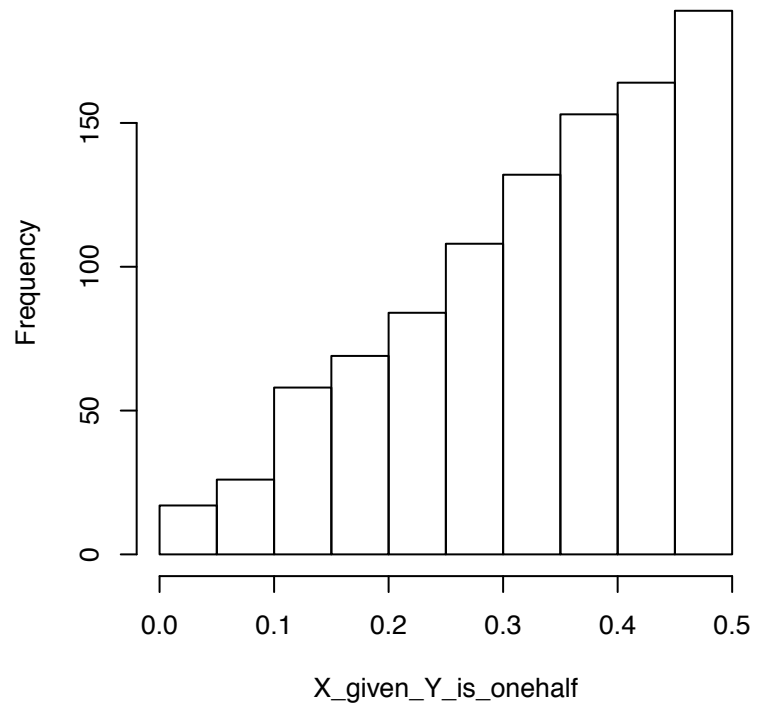


Joint Distribution of Weights of Different Nuts

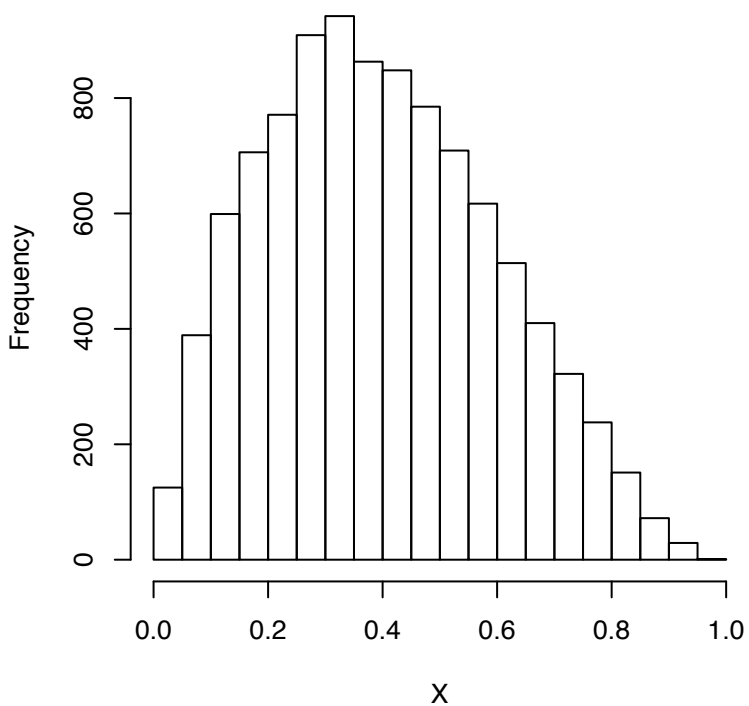
Joint Samples of (X, Y)



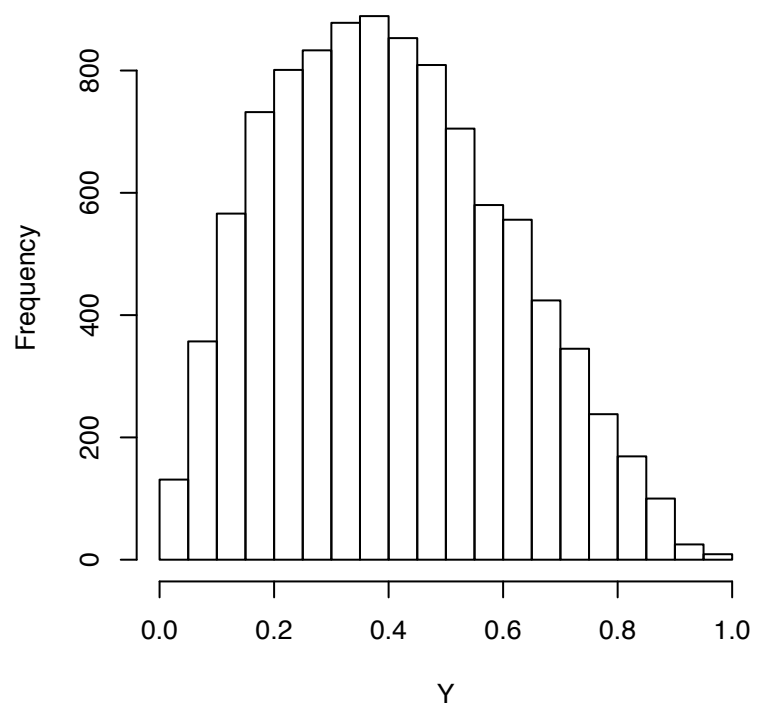
Histogram of X_given_Y_is_onehalf



Histogram of X



Histogram of Y



Bivariate Normal Distributions

Section 5.3

Probability Density Functions

The PDF of a bivariate Normal distribution parameterized by $\mu_x, \mu_y, \sigma_x, \sigma_y$, and ρ is given by:

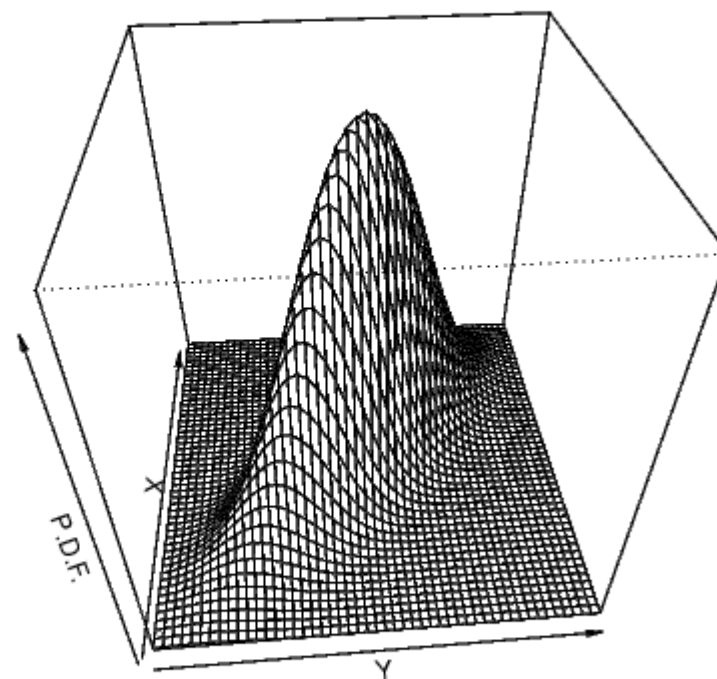
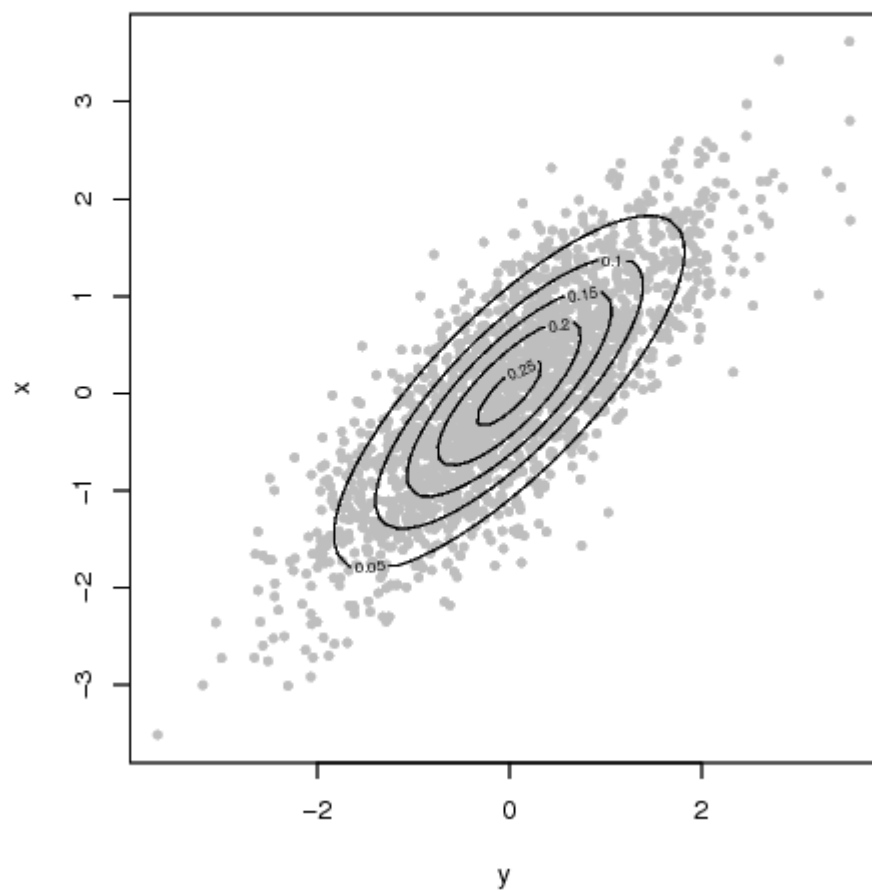
$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_x}{\sigma_x} \right)^2 + \left(\frac{y-\mu_y}{\sigma_y} \right)^2 - 2\rho \left(\frac{x-\mu_x}{\sigma_x} \right) \left(\frac{y-\mu_y}{\sigma_y} \right) \right] \right]$$

The marginal distribution of X and Y are respectively:

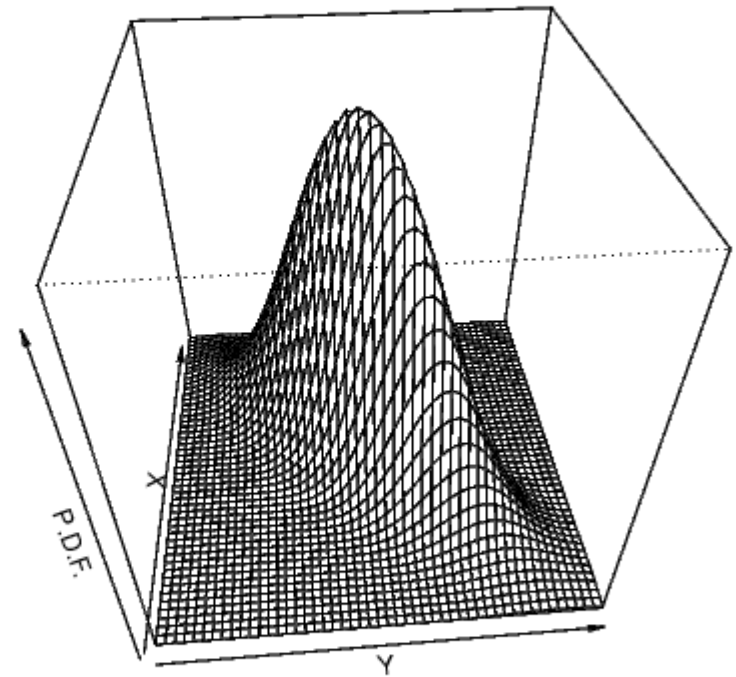
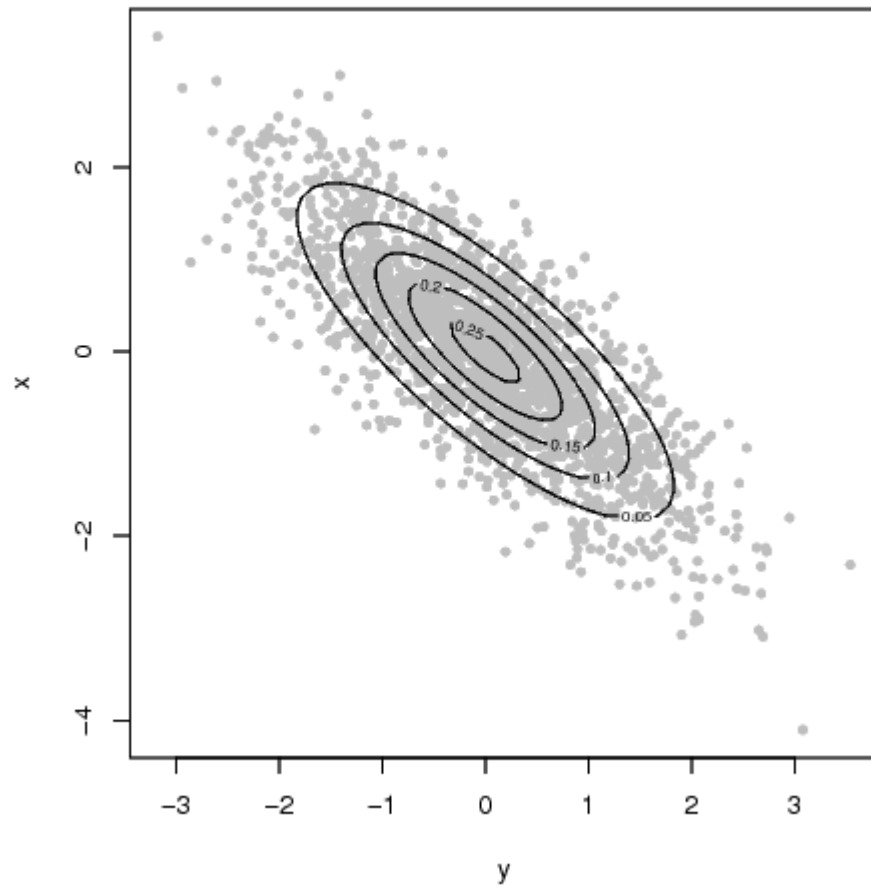
$$X \sim N(\mu_x, \sigma_x^2), \quad Y \sim N(\mu_y, \sigma_y^2)$$

Visualizing Bivariate Normal Distributions

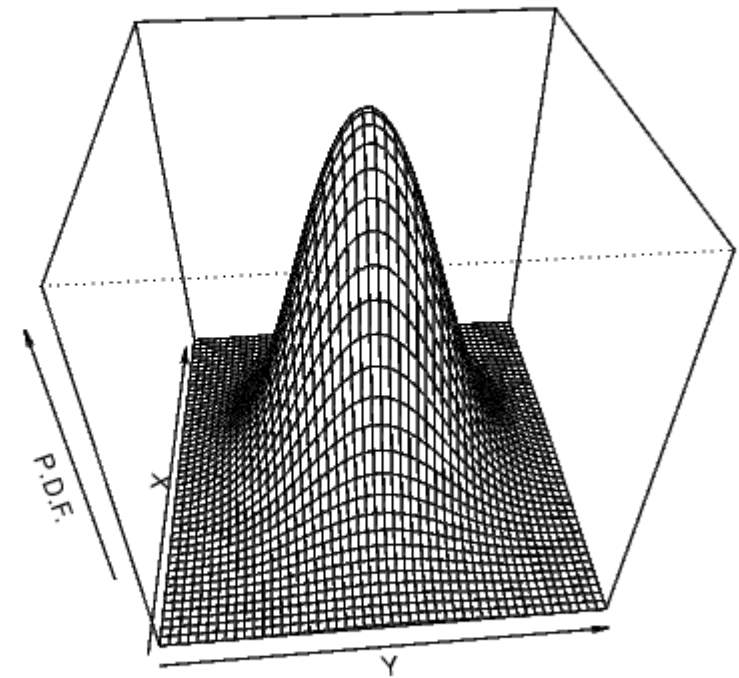
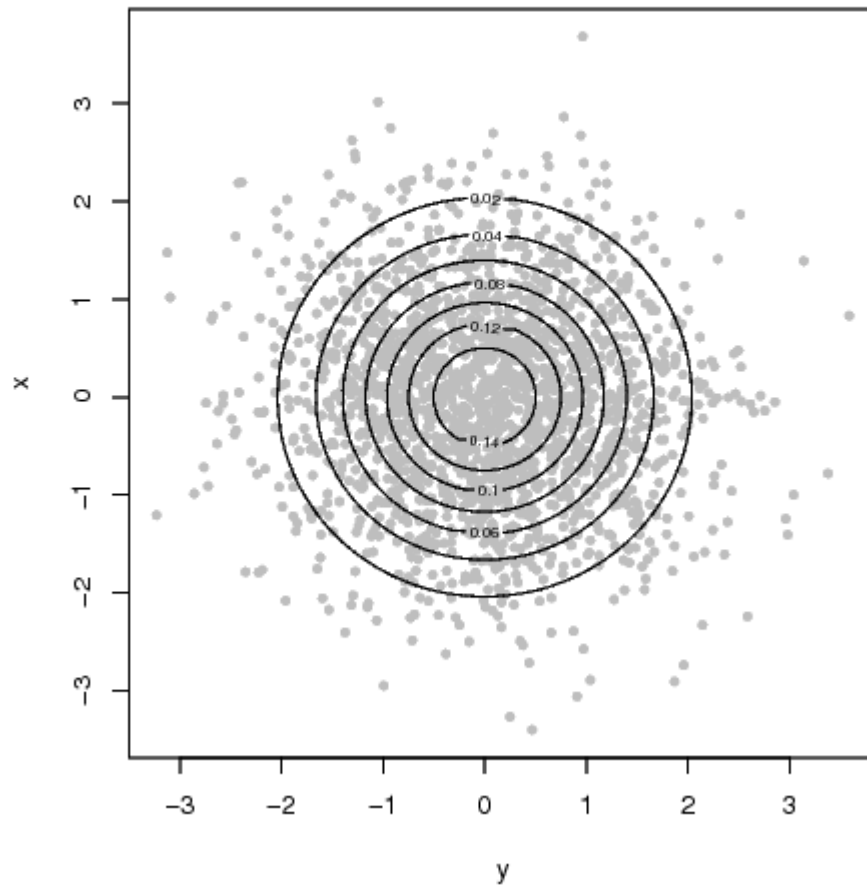
Bivariate Normal Distribution with $\rho = 0.8$



Bivariate Normal Distribution with $\rho = -0.8$



Bivariate Normal Distribution with $\rho = 0$

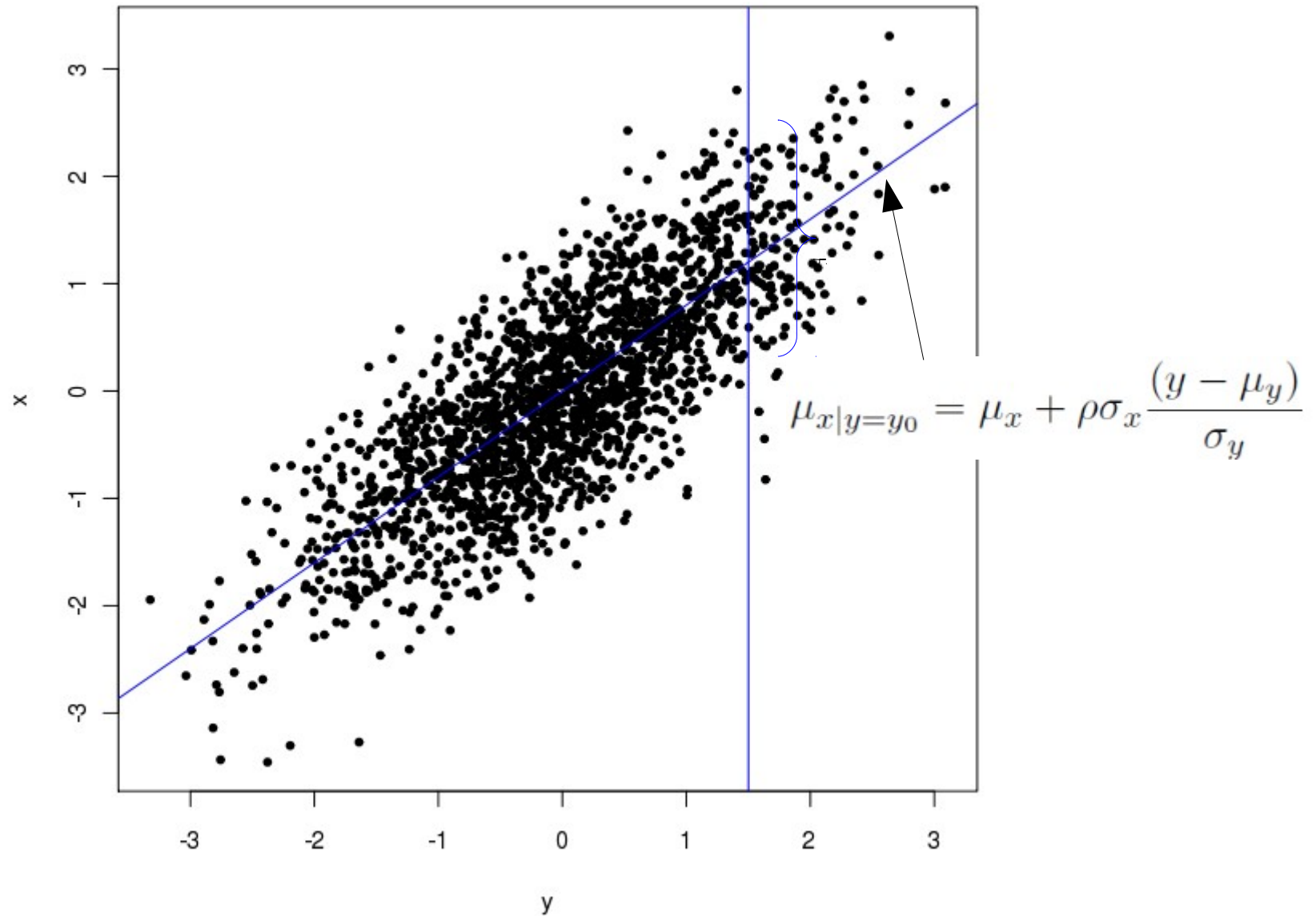


Conditional Distributions

A powerful feature of the bivariate normal distribution is that the conditional probability distribution function for one of the variables, given a known value for the other variable, is normally distributed with $N(\mu_{x|y=y_0}, \sigma_{x|y=y_0}^2)$, where

$$\mu_{x|y=y_0} = \mu_x + \rho\sigma_x \frac{(y - \mu_y)}{\sigma_y}$$
$$\sigma_{x|y=y_0} = \sigma_x \sqrt{1 - \rho^2}$$

Graphs of Conditional Distribution - samples



Graphs of Conditional Distribution - density

